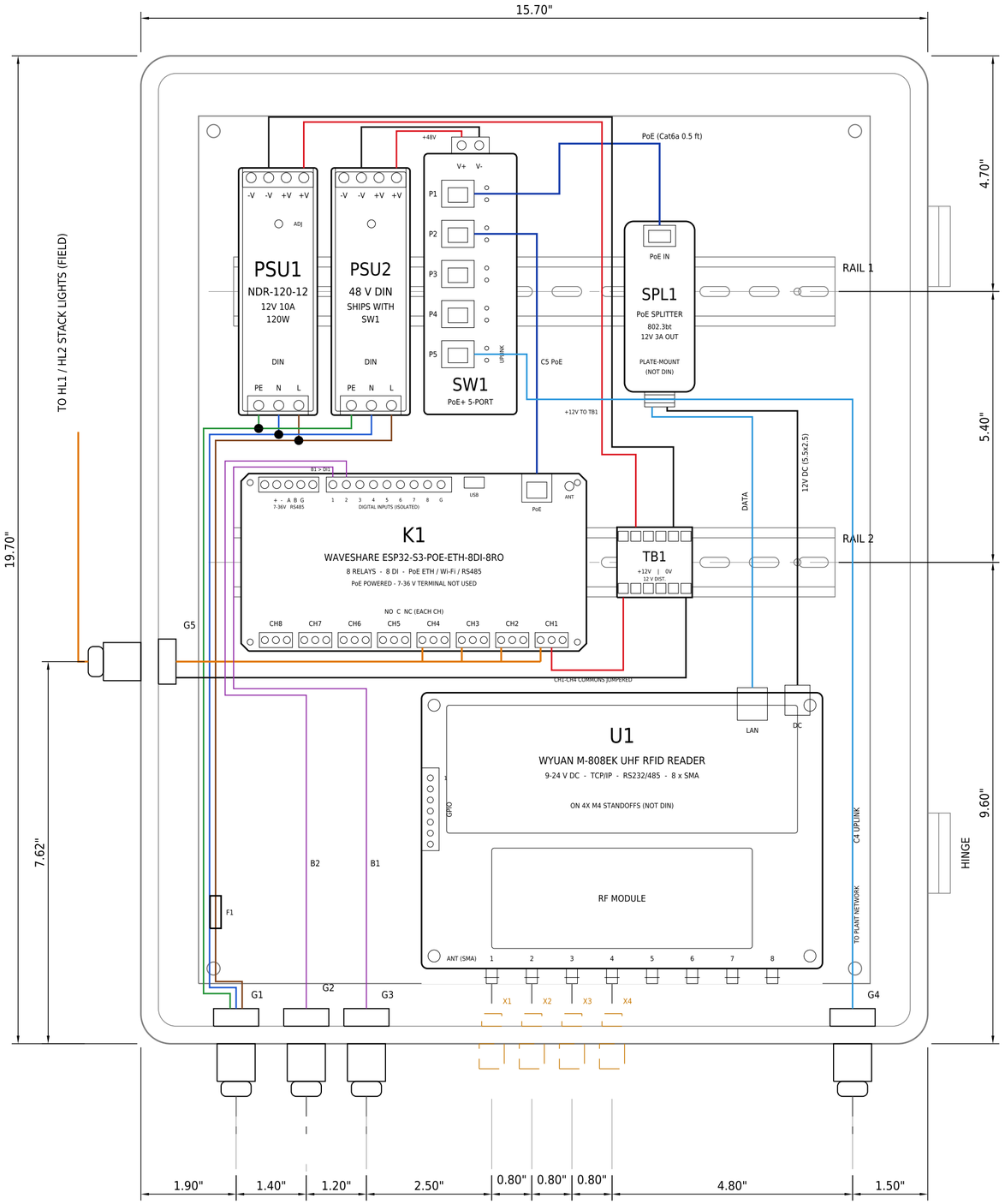
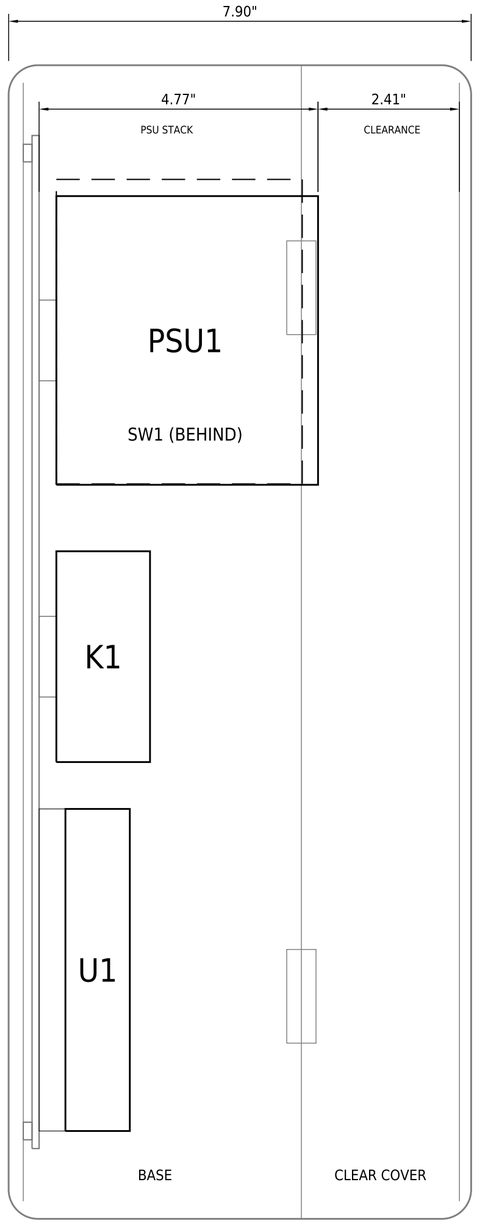
REV	DESCRIPTION	DATE
6	K1 > 8DI-8RO PoE BD; B1/B2 TO K1 DI; B1 DIHOOL, B2 IRR-7R; TB1 ADDED	2026-10-17
7	PSU2 48 V FOR SW1 (SW1/SPL1 SHIFTED); B2 = IRR-7M AS ORDERED	2026-10-17
8	C4 UPLINK VIA G4 TO SW1 P5 AND F1 MADE REQUIRED; WIRE 203 ADDED; MATCHES PARTS LIST REV 5	2026-10-17



FRONT VIEW - COVER REMOVED

SCALE 1:2



RIGHT SIDE VIEW

SCALE 1:2

DEVICE ENVELOPES ONLY

PARTS LIST

ITEM	REF	QTY	DESCRIPTION
1	PSU1	1	NDR-120-12 DIN PSU 12V 10A 120W
2	SW1,PSU2	1	GOODTOP PoE+ 5-PORT + 48V DIN PSU
3	SPL1	1	REVODATA PS5712BG-25 PoE SPLITTER
4	K1	1	WAVESHARE ESP32-S3-POE-ETH-8DI-8RO
5	U1	1	WYUAN M-808EK UHF READER, 8 SMA
6	-	3	DIN RAIL 35 x 7.5 mm, 12.00" (1 SPARE)
7	B1	1	DIHOOL M16 LASER, NPN NO (FIELD)
8	B2	1	JUTAI IRR-7M BEAM + REFL. (FIELD)
9	HL1,2	2	STACK LIGHT R/A/G + BUZZER (FIELD)
10	-	1	TICONN IP67 ENCL. 19.7x15.7x7.9
11	W1	2	CAT6A PATCH 0.5 FT (SPL1, K1)
12	G1-G5	5	1/2" NPT CABLE GLAND
13	X1-X4	4	BNC BULKHEAD + RG316 TO SMA*
14	TB1,F1	1	12V TERMINALS 6-POS + F1 FUSE HLDR

* RECOMMENDED - NOT ON BUILD LIST

NOTES:

1. DIMENSIONS IN INCHES. GEOMETRY IS DRAWN 1:1 IN MODEL SPACE; SHEET FRAMES ARE SIZED TO PLOT AT 1:2 ON ANSI C (22 x 17).
2. MOUNTING PLATE ASSUMED 13.4 x 17.3 - VERIFY AGAINST THE TICONN PLATE BEFORE DRILLING OR CUTTING RAILS.
3. DEVICE OUTLINES ARE ENVELOPES. NDR-120-12 IS 40 x 125.2 x 113.5 mm PER DATASHEET; OTHER SIZES ARE APPROXIMATE - VERIFY.
4. SPL1 AND U1 ARE NOT DIN-MOUNT: SPL1 ON A BRACKET / TIE BASE, U1 ON 4X M4 STANDOFFS.
5. KEEP AC (L / N / PE) SEPARATE FROM ETHERNET AND SIGNAL WIRING.
6. DASHED ORANGE = RECOMMENDED, NOT ON BUILD LIST. DASHED BLACK = FIELD OR HIDDEN.
7. ROUTING IS INDICATIVE. SEE SHEET 2 FOR TERMINATIONS AND WIRE NUMBERS.
8. G1 AC IN, G2 SAFETY BEAM (B2), G3 TRUCK SENSOR (B1), G4 NETWORK UPLINK, G5 STACK LIGHTS. X1-X4 = DOCK-DOOR ANTENNAS 1-4.
9. K1 IS PoE-POWERED FROM SW1 P2 (C5). TB1 FEEDS 12 V / 0 V TO RELAY COMMONS, B1, B2 AND THE STACK LIGHTS (SENSOR SUPPLY LEADS NOT SHOWN).
10. K1 IS 175 x 90 mm PER WAVESHARE - VERIFY FIT ON RAIL 2 BEFORE CUTTING.
11. SW1 IS POWERED BY PSU2 (48 V DIN SUPPLY THAT SHIPS WITH SW1). F1 / G1 AC FEEDS PSU1 AND PSU2.
12. C4 UPLINK: CAT6 FROM THE PLANT SWITCH THROUGH G4 TO SW1 P5. F1 = DIN FUSE HOLDER ON THE L CONDUCTOR.

WIRE / CABLE LEGEND

- AC LINE (L)
- AC NEUTRAL (N)
- PROTECTIVE EARTH
- +12 / +48 V DC
- 0 V / DC (-)
- PoE (DATA + POWER)
- ETHERNET DATA
- SENSOR CABLE
- STACK-LIGHT CABLE
- ANTENNA COAX
- RECOMMENDED / RESERVED

GIELOW PICKLES · G5 CROSWELL

INVENTORY CONTROL / AUTOMATION

TITLE

RFID PANEL BOX

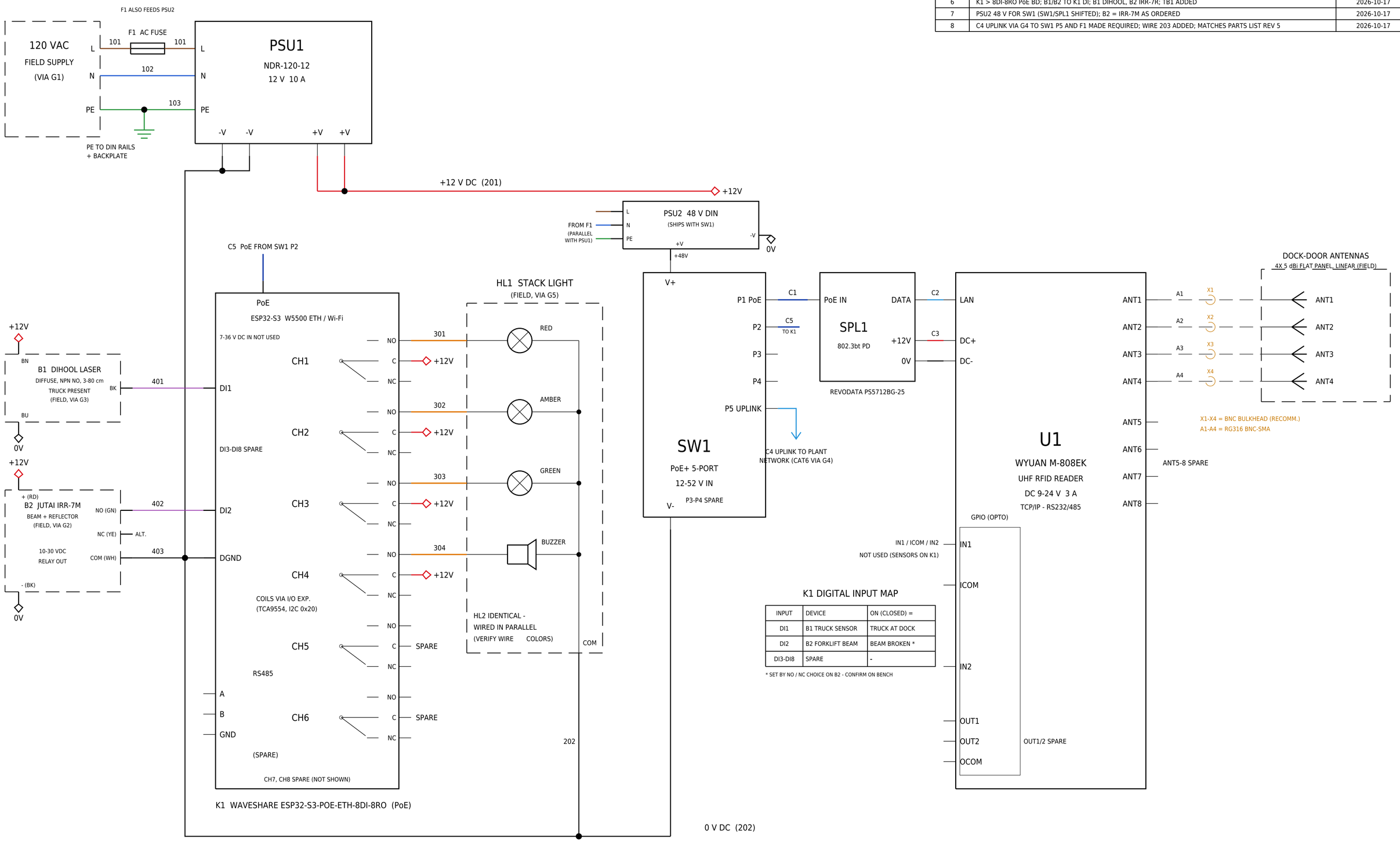
PANEL LAYOUT - FRONT & SIDE VIEWS

DWG NO.	SIZE	REV	SHEET
RPB-001	C	8	1 OF 2

SCALE:	UNITS:	DATE:
1:2	INCHES	2026-10-17

DRAWN:	CHECKED:	APPROVED:

REV	DESCRIPTION	DATE
6	K1 > 8DI-8RO PoE BD; B1/B2 TO K1 DI; B1 DIHOOL, B2 IRR-7R; TB1 ADDED	2026-10-17
7	PSU2 48 V FOR SW1 (SW1/SPL1 SHIFTED); B2 = IRR-7M AS ORDERED	2026-10-17
8	C4 UPLINK VIA G4 TO SW1 P5 AND F1 MADE REQUIRED; WIRE 203 ADDED; MATCHES PARTS LIST REV 5	2026-10-17



K1 DIGITAL INPUT MAP

INPUT	DEVICE	ON (CLOSED) =
DI1	B1 TRUCK SENSOR	TRUCK AT DOCK
DI2	B2 FORKLIFT BEAM	BEAM BROKEN *
DI3-DI8	SPARE	-

* SET BY NO / NC CHOICE ON B2 - CONFIRM ON BENCH

WIRE / CABLE LIST

NO.	SIGNAL	COLOR	TYPE
101	AC LINE (L)	BROWN	18 AWG
102	AC NEUTRAL	BLUE	18 AWG
103	PROT. EARTH	GN/YE	18 AWG
201	+12 V DC	RED	18 AWG
202	0 V DC	BLACK	18 AWG
203	+48 V DC (PSU2>SW1)	RED	18 AWG
301-304	STACK LIGHT	PER TOWER	18 AWG 5C
401-403	K1 DI INPUTS	BK / GN / WH	22 AWG
C1, C5	SW1 > SPL1, K1	BLUE	CAT6A 0.5 FT
C2 / C3	SPL1 > U1	-	PIGTAILS
C4	UPLINK > SW1 P5	-	CAT6 (FIELD)
A1-A4	ANTENNA	-	RG316 COAX

RELAY MAP / STACK-LIGHT LOGIC

RELAY	OUTPUT	DOCK STATE
CH1	RED	NO TRUCK AT DOCK
CH2	AMBER	TRUCK DOCKED - READER IDLE UNTIL BEAM
CH3	GREEN	FORKLIFT LOADING TRUCK
CH4	BUZZER	ALERT (OPTIONAL)
CH5-CH8	SPARE	-

NOTES:

1. ALL RELAY COMMONS (C) JUMPERED TO +12 V (201); NO CONTACTS SWITCH +12 V TO EACH LAMP.
2. K1 IS PoE-POWERED FROM SW1 P2 (C5). DO NOT ALSO FEED 7-36 V. F1 FEEDS BOTH PSUs.
3. B1 IS NPN (SINKING): BK TO DI1. B2 IS A DRY CONTACT: NO (OR NC) TO DI2, COM TO DGND.
4. K1 DGND BONDED TO 0 V (202) SO PSU1-FED SENSORS SHARE A REFERENCE - VERIFY AGAINST WAVESHARE DI WIRING DIAGRAM.
5. U1 IS POWERED FROM SPL1 (ISOLATED PoE OUTPUT); ITS DC- IS NOT BONDED TO THE 0 V BUS. U1 GPIO NOT USED.
6. SW1 RUNS FROM PSU2 (48 V, SHIPS WITH SW1); PoE ON P1 AND P2.
7. VERIFY STACK-LIGHT WIRE COLORS WITH MANUFACTURER DATA.

GIELOW PICKLES · G5 CROSWELL
INVENTORY CONTROL / AUTOMATION

TITLE
RFID PANEL BOX
WIRING SCHEMATIC

DWG NO.	SIZE	REV	SHEET
RPB-001	C	8	2 OF 2

SCALE:	UNITS:	DATE:
NONE	INCHES	2026-10-17

DRAWN:	CHECKED:	APPROVED: